Overview

Our user research helped us understand how UQ students collaborate during group assessments and what makes these experiences easier or more difficult. We focused on how students communicate, share work, contribute to group tasks and move between working together in person and continuing their work online.
Initially, we used interviews to learn about students' existing experiences with group work. Following this, we conducted a workshop to determine how students' interacted with butchers paper when given a task. We then ran a workshop to explore how physical figurines could fit into a collaborative activity and whether they could help connect individual users to a shared workspace.
The findings from these activities influenced the development of Zook, particularly our decision to combine a shared digital workspace with individual user identities and physical figurines.

1. Interviews

Approach

Each member of our group conducted interviews using a semi-structured interview approach. We asked students about their previous experiences with in-class group assessments, the tools they used to collaborate, how they organised their work, and what made a collaborative environment effective.
The interviews explored questions such as:

- I want you to think about when you last collaborated on an in-class activity with a team and tell me how it went/what you did. (Follow up: Did you use any physical means to communicate?)
- How would you define an effective collaborative space for your team? Is there ever a time where you think you have worked in an ideal collaborative space for team work? tell us about it.
- Tell me about your last in class team-work experience - How did you transfer in-class activities and collaboration to external team work?
- Think about your last in-class activity - tell me your experience about initially setting up a group collaboration tool? (Why did you pick that tool over others? Did you have a choice?)
- Walk me through how your group usually decides on their collaboration tools, what kept you engaged and what do you think could have been better (Follow on: what would have kept you more engaged?)
- Think about a recent team assessment. How did you keep track of what each person contributed or was working on? (Follow up: Was work ever duplicated, missed or the workload felt unfair? How was it handled?)

- During in-class group work, how do you make sure everyone has a chance to contribute ideas or be involved in decisions? (Follow up: Have you or a team member ever felt left out or less involved? What do you think was the cause?)

The questions were intended to encourage participants to describe their own experiences rather than simply respond to our ideas about what a collaboration tool should do.

Key findings

Moving between physical and digital work

Students often moved between face-to-face discussions and digital tools. They might brainstorm ideas together in class, then use group chats, shared documents, presentations or other platforms to continue working outside the classroom. This highlighted the importance of supporting the transition between these different stages of collaboration.

Fair contribution and accountability

Students valued having a clear role within the group and wanted to feel included in the process. However, having a group discussion did not necessarily mean everyone contributed equally to the actual work. We therefore began exploring how individual participation could be supported without taking away the shared nature of the activity.

Awareness of teammates

Students wanted to understand what other group members were doing so they could coordinate tasks, avoid duplicated work, and reduce confusion. This suggested that collaboration involves more than providing a space to create content. Students also need to stay aware of each other's contributions and progress.

The physical environment matters

The interviews also highlighted the importance of the space in which collaboration takes place. Participants described environments that helped them focus and communicate, and one participant suggested that a circular table could help everyone focus on the group. This supported our interest in a shared tabletop rather than a design that required everyone to work independently on separate screens.

Different levels of confidence with technology

Students had different levels of confidence with digital collaboration tools. We therefore needed to consider how Zook could be easy to understand and use, rather than introducing another complicated system that groups would need to learn.

How the findings influenced Zook

The interviews helped us recognise that the problem was not simply a lack of collaboration tools. Students already had access to many tools, but their work was often spread across physical activities and separate digital platforms.
This led us to explore a shared digital workspace that could preserve the social experience of working together while giving individuals their own identity and controls. We initially considered using UQ ID cards to establish a shared workspace, but later explored figurines as a more physical and playful way for users to connect with the system.

2. Workshop One: Creating a Book Cover and Story

Activity

In our first workshop, participants worked together on a piece of butcher paper to create a book cover and develop a story. We introduced physical figurines as representations of the individual participants.
The purpose was to observe how participants interacted with the figurines during a collaborative activity and whether they naturally understood them as representations of themselves.

What we observed

Participants mainly treated the figurines as characters or objects within the story they were creating. Rather than consistently using them to represent their own identities, they incorporated them into the activity itself.
This was an important finding because we had initially assumed that participants would naturally associate a figurine with themselves. The workshop showed that this connection was not automatic.

What we learned

The first workshop challenged our assumption that simply providing each participant with a physical object would establish a connection between that person and the digital workspace.
It also showed that the activity itself influenced how the figurines were interpreted. Because participants were creating a story, it made sense for them to treat the figurines as story characters.
This led us to reconsider how the figurines should be introduced and used in our next workshop.

3. Workshop Two: Designing a Dream Car

Activity

For the second workshop, we changed the activity and asked participants to collaboratively draw and describe their dream car. This time, participants were required to hold their figurine when they wanted to contribute to the page.
Unlike the first workshop, the figurines were intended to represent the participants more directly, rather than act as characters within a story. We wanted to explore whether physically holding a figurine would strengthen the connection between the participant and their contribution.

What we observed

Participants showed interest in the figurines and often examined, handled, or played with them. The figurines attracted attention and sometimes appeared to make the activity more engaging. However, participants did not always remember to pick up their figurines when contributing. Teammates sometimes had to remind each other to hold them. One participant placed their figurine on the paper and wrote with their other hand, suggesting that keeping the figurine in their hand was inconvenient.
We also observed that the physical contribution to the activity was not evenly distributed. Often, only one or two participants were actively writing or drawing at a time, while others contributed through discussion or allowed teammates to lead. Some participants needed prompting to encourage others to contribute.

What we learned

The second workshop showed that interest in a playful object does not necessarily translate into more balanced teamwork. Although participants were curious about the figurines, requiring them to hold one whenever they contributed created an extra step that could interrupt the activity.
We also recognised that we should not treat physical interaction as the only meaningful form of participation. Discussing ideas and responding to teammates can also contribute to collaboration, even when someone is not directly interacting with the workspace.
The findings suggested that the figurines needed to be convenient to use and integrated more naturally into the shared environment.

5. Limitations and Considerations

Our research provided useful early insights, but the findings should be interpreted within the context of the activities we conducted.
The workshops involved specific tasks, and the book-cover activity may have influenced participants to treat figurines as story characters. Similarly, requiring participants to hold

figurines during the dream-car activity introduced a physical constraint that may not be necessary in the final design.
The workshops also showed that playful interactions alone cannot guarantee equal participation. Group dynamics, confidence, task roles, and the willingness to invite others into a discussion may continue to influence how people collaborate.
For these reasons, we treated the workshops as opportunities to identify design problems and refine our ideas, rather than as proof that figurines would automatically improve teamwork.

6. Interview Scripts and Supporting Materials

The interview scripts used during our research are included in our project documentation. They provide the original questions used to explore students' experiences with group assessments, collaboration tools, communication, and participation.
Supporting materials: Interview Scripts and Interview Notes - Where Taken
Workshop activity documentation: Prototype and Workshop Images
These materials provide additional context for the findings and show how our design decisions developed from the research.